Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

SST TAC Production Centre

Baltic Sea - Diurnal Subskin Sea Surface Temperature Analysis

SST_BAL_PHY_SUBSKIN_L4_NRT_010_034

Issue: 2.0

Contributors: I. Karagali, W. Kolbe , J. Høyer.

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	June 2022	All	Creation of the document	I. Karagali, W. Kolbe, J. Høyer	A. Pisano
2.0	May 2025	All	Updated template, inclusion of VIIRS sensor on NOAA21.	Ioanna Karagali, Jacob Høyer, Matilde B. Kreiner	A. Pisano

ACRONYM TABLE

Acronym	Description
AVHRR	Advanced Very High Resolution Radiometer
Argo	International profiling float network (www.argo.net)
DMI	Danish Meteorological Institute
EiS	Entry into Service
ESA	European Space Agency
EUMETSAT	European Meteorological Satellites
GHRSST	Group for High Resolution Sea Surface Temperature
MetOp	Meteorological Operational Satellites
MSG	Meteosat Second Generation
NASA	National Aeronautics and Space Administration
NOAA	National Oceanic and Atmospheric Administration
NPP	National Polar-orbiting Partnership
NRT	Near Real Time
OI	Optimal Interpolation
SAF	Satellite Application Facility
SEVIRI	Spinning Enhanced Visible and Infrared Imager
SIC	Sea Ice Concentration
SLSTR	Sea and Land Surface Temperature Radiometer
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre
VIIRS	Visible Infrared Imaging Radiometer Suite

KEY DEFINITIONS

Level-2 Pre-processed (L2P) data product	Geophysical variables derived from Level 1 source data on the Level 1 grid (typically the satellite swath projection). They are ancillary data and metadata added following GHRSST Data Specification.
Level-3 Uncollated (L3U) data product	L2 data granules remapped to a regular latitude/longitude grid without combining observations from multiple source files. L3U files will typically be “sparse”, corresponding to a single satellite orbit.
Level-3 Collated (L3C) data product	SST observations from a single instrument combined into a space-time grid. In this project, a typical L3C file may contain all the observations from a single instrument in a 24-hour period.
Level-3 Super-Collated (L3S) data product	SST observations from many instruments combined into a space-time grid. In this project, a typical L3S file may contain all the observations from all available instruments in a 24-hour period.
Level 4 (L4) data product	SST observations from multiple instruments using an analysis system (e.g. optimal interpolation) to produce a gridded, gap-free product.
Sea Ice	Corresponds to ocean surface covered by sea ice (>70%) thus the surface temperature is classified as sea-ice surface temperature (IST).
Sea Ice Concentration	Fraction (percentage) of the sea surface area of the grid cell containing sea-ice.
Sea Surface Temperature (SST)	SST is the temperature of the ocean water close to the surface.
Sea Surface Temperature at the base of the thermal layer ($SST_{foundation}$)	The sea surface foundation temperature ($SST_{foundation}$) represents the average temperature of the upper ocean layer, free from diurnal (daily) fluctuations, such as those caused by solar heating.
Sea Surface Temperature at the base of the skin layer ($SST_{sub\text{-}skin}$)	The sea surface subskin temperature ($SST_{sub\text{-}skin}$) is the sea surface temperature at the base of the ocean's "skin layer," a very thin (less than 1 mm) layer where heat transfer is dominated by conduction, not turbulence.
Thermal skin layer	The layer within which the long-wave radiation penetrates the ocean, typically less than 1mm, where ocean-atmosphere heat exchange takes place and temperature is measured using Infra-red sensors.

RELEVANT DOCUMENT LIST

CMEMS-SST-SQO-010-034	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated product page[1] under Documentation[2]
CMEMS-SST-PUM-010-034	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page under Documentation[3]
CMEMS-SST-SQO-010-007-032	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated product page[4] under Documentation[5]
CMEMS-SST-PUM-010-007-032	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page under Documentation[6]

[1]https://data.marine.copernicus.eu/product/SST_BAL_PHY_SUBSKIN_L4_NRT_010_034/description
[2]https://documentation.marine.copernicus.eu/SQO/CMEMS-SST-SQO-010-034.pdf
[3]https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-034.pdf
[4]https://data.marine.copernicus.eu/product/SST_BAL_SST_L4_NRT_OBSERVATIONS_010_007_b/description
[5]https://documentation.marine.copernicus.eu/SQO/CMEMS-SST-SQO-010-007-032.pdf
[6]https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-007-032.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The product validated in this document is the level 4 (L4) Diurnal Sub-skin Sea Surface Temperature Analysis **SST_BAL_PHY_SUBSKIN_L4_NRT_010_034** available hourly (May 1st, 2021 - August 31st, 2021) at 0.02° resolution for the Baltic Sea and North Sea, produced by the Danish Meteorological Institute (DMI). The DMI L4 Diurnal Sub-skin Sea Surface Temperature Analysis consist of hourly, gap-free fields of sea surface temperature and sea ice area fraction; referred to as L4 product at 0.02° x 0.02° horizontal resolution, using infra-red satellite observations and high-resolution sea ice information from SMHI and the Copernicus Marine Service Sea Ice TAC.

I.2 Summary of the results

The SST product is compared to in-situ SST measurements from drifting buoys and difference statistics are derived in terms of mean bias to the buoys. The results are summarised below.

Temperature: The bias between the L4 Diurnal SST product and drifting buoys available is -0.17°C while when moored buoys are used, the mean bias is 0.05°C. The corresponding standard deviation values are 0.55°C and 0.80°C respectively. The reason the latter is higher is due to the more coastal locations of the moored buoys and the fact that they often indicate the temperature at 0.5 – 2m depth.

Important information or **noticeable event**: In the course of the operational system analysis, a new sensor (i.e. VIIRS on NOAA21) has been introduced.

I.3 Estimated Accuracy Numbers

A summary of the estimated accuracy numbers, derived from the validation (see sections VI.1 and VI.2) of the L4 product against drifting buoys (top row) and moored buoys (bottom row) is presented in Table 1.

Table 1. Table of estimated accuracy numbers (where variables, units and numbers are given).

Buoy type	Variable	Units	Metrics	"Values of metric"
Drifting	Temperature	°C	mean bias	-0.17
			standard deviation	0.55
			Root mean square deviation	0.58
			number of observations (Nobs)	346990
Moored			mean bias	0.05
			standard deviation	0.80
			Root mean square deviation	0.81
			number of observations (Nobs)	86317

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: Danish Meteorological Institute, Denmark (DK)

Production system name: DMI L4 Diurnal Sub-skin Sea Surface Temperature Analysis (CMEMS name: SST_BAL_PHY_SUBSKIN_L4_NRT_010_034 – cmems-obs-sst-bal-phy-subskin-nrt-l4_P1H-m)

Description: The Danish Meteorological Institute Optimal Interpolation (DMIOI) analysis system is run daily in the operational suite at DMI. It is run at 06:00 UTC and is completed before 09:30 UTC for the 24-hourly fields of the previous day. Products are typically available to users starting at 06:30 UTC and are updated frequently as the hourly analysis is completed.

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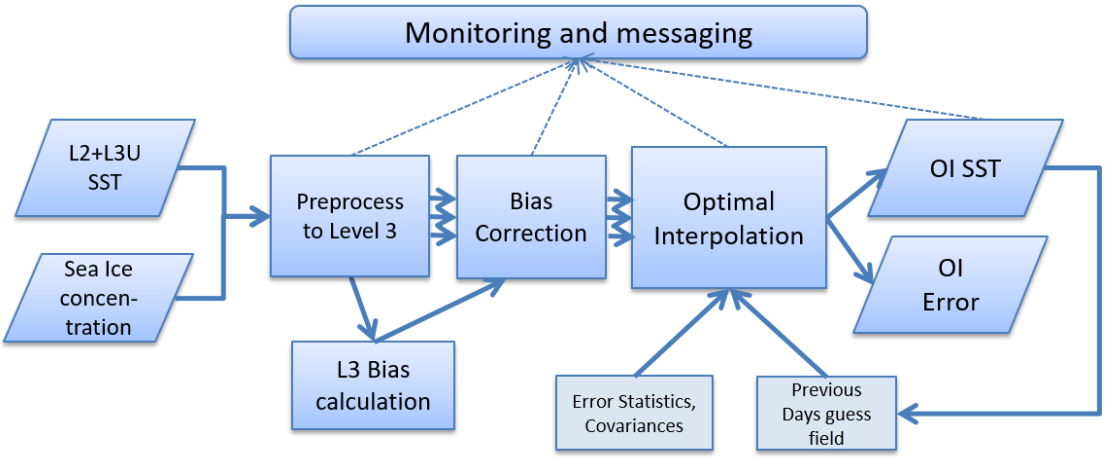

Figure 1. Schematic diagram of the operational processing chain of the Optimal Interpolation (OI) system at Danish Meteorological Institute (DMI) used for the L4 Diurnal SST product.

The SST system running at DMI uses a persistence approach based on the use of the previous analysis field as a first guess (see Figure 1). The SST observations from the current day are therefore interpreted as anomalies with respect to the first guess field. The errors on the guess field are derived from the data. The pre-processing steps consist of the hourly collation of each of the different products on a L3 grid. The time scale for the aggregation is the analysis hour ± 30 minutes. For more information on the processing system and the Optimal Interpolation (OI) method see the Product User Manual[7], Høyer & She (2007) and Høyer & Karagali (2016). For more information on diurnal variability in the North Sea/Baltic Sea region see Karagali et al. (2012) and Karagali & Høyer (2014).

[7] See https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-034.pdf

II.1 Quality control and pre-processing of input data

All satellite data valid for a particular day and hour (i.e., within 24 hours from the analysis) are considered. The input L2/L3U SST data undergo various quality control (QC) and processing steps to generate separate level 3 products:

- Only satellite SST data, which have a quality flag of 4 and higher are accepted (quality flag levels: 0 = no_data, 1 = bad data, 2 = worst quality, 3 = low quality, 4 = acceptable quality, 5 = best quality)..
- Single-sensor hourly files are bias-corrected against the reference sensor to ensure consistency between all input SST.

Hourly anomalies (dSST) from the SST foundation temperature of the same day are estimated as SST_{hour} minus $SST_{foundation}$. The optimal interpolation (OI) is then applied to the hourly anomaly field to produce gap-free hourly dSST and, the $SST_{foundation}$ is subsequently added to the hourly dSST.

The main SST analysis uses a persistence approach based on the use of the previous analysis field as a first guess. The supercollated L3 (L3S) SST hourly observations from the current day are interpreted as anomalies with respect to the first guess field; and local statistics have been included to estimate the optimal SST values for the final product[8], see Høyer & She (2007) and Høyer & Karagali (2016).

II.2 Input data

The data are collected in near real-time (NRT) from various satellite SST producers (NASA, NOAA, EUMETSAT SAF on Ocean and Sea Ice) in GHRSST Level 2 (L2P) format (i.e., in satellite swath coordinates). Specifically, the following satellite SST level 2 data are collected as inputs for the DMIOI algorithm:

- **Infra-red data** from the Advanced Very High Resolution Radiometer (AVHRR) instruments on board NOAA (National Oceanographic and Atmospheric Administration) and MetOp (Meteorological Operational) satellites;
- **Spinning Enhanced Visible and Infrared Imager** (SEVIRI) on board the Meteosat Second Generation (MSG) satellite;
- **Visible Infrared Imaging Radiometer Suite** (VIIRS) on board the Suomi-NPP (Suomi National Polar-orbiting Partnership), NOAA20 and NOAA21;
- **Sea and Land Surface Temperature Radiometer** (SLSTR) on board the Sentinel 3-A/B satellites.

The analysis also uses high resolution (1 km) **sea ice concentration products over the Baltic Sea** from the Copernicus Marine Service Sea Ice (SI) TAC (Karvonen & Simila, 2007), and for operational validation of in-situ SST measurements from drifting and moored buoys; obtained from the CMEMS In Situ Thematic Assembly Centre (INS TAC). The ice fields from the sea ice concentration products are used as a mask, where sea ice concentrations higher than 30% have been set to a SST of -0.8°C, to represent the freezing temperature in the brakish waters in the Bay of Bothnia (see e.g. Høyer & Karagali, 2016).

[8] See also the Product User Manual https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-034.pdf

III VALIDATION FRAMEWORK

The SST L4 hourly product covers the North Sea and Baltic Sea. The validation of SST TAC products is based on the use of drifting buoy measurements exclusively[9], as recommended currently by the GHRSST group on satellite SST validation. In-situ observations are not included in the hourly analysis product and the validation statistics therefore represent an independent estimate of the performance of the L4 SST hourly analysis. The mean and standard deviation of the departures from drifting buoy SST measurements are computed over pre-defined geographical areas, such as the full domain; (including both the Baltic Sea and North Sea basins), the Baltic Sea only and the North Sea. The longitude of 9ºE divides into Baltic Sea (East of 9ºE) and North Sea (west of 9ºE).

Various quality check procedures are applied on drifting buoy measurements by the provider to discard suspect data (that result from e.g. sensor malfunction, deployment/operational issues, environmental factors, or errors during data processing and transmission), using gross error checks and/or blacklists(e.g., lists of buoys with erroneous recordings, made available by various centres)[10]. These checks identify data flagged as suspect.

The lack of drifting buoy observations in the Baltic Sea and in large parts of the North Sea, makes it necessary to supplement the drifting buoy observations with in-situ observations from moored buoys (Figure 2). Observations from drifting and moored buoys are obtained from the Copernicus Marine Service In-Situ TAC[11]. As the observations provided by these different types of buoys also have different error characteristics (e.g., drifting buoys typically observe at 20 cm depths, whereas moored buoys can be several meters depths and often coastal), all the validation statistics shown here will be separated into, drifting and moored buoys when available. Note that the coastal locations of the buoys will likely result in higher uncertainties compared a spatially representative in-situ data set.

[9] See https://www.ghrsst.org/about-ghrsst/task-teams/hrsst-drifters-for-satellite-sst-validation/
[10] See e.g. https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-030-036.pdf
[11] See https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/description

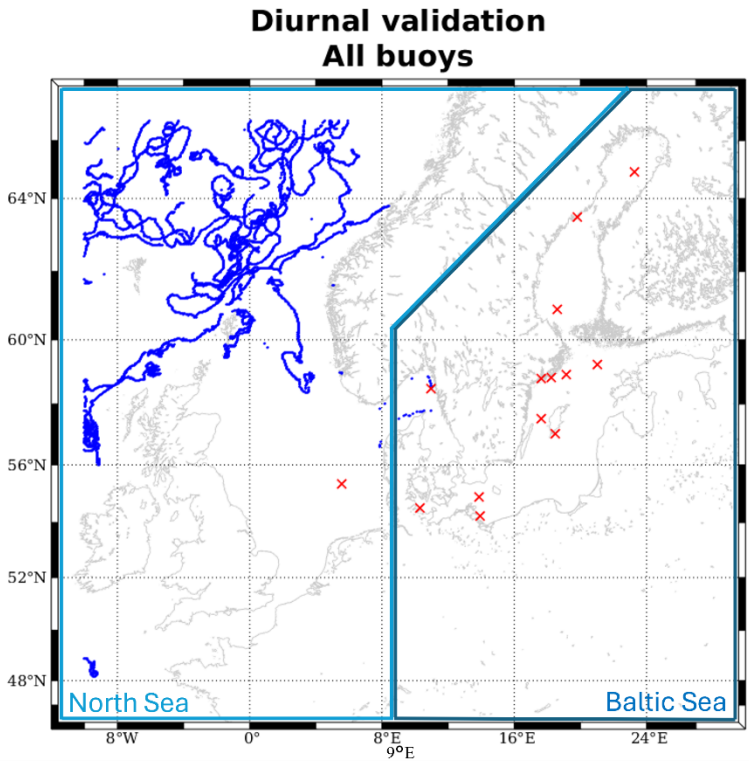

Figure 2. Overview of drifting buoy observations (cobolt blue lines) and moored buoy locations (red crosses) used for validation (May 1st, 2021- Aug 31st, 2021). Also shown here are the comparison subregions North Sea (Turquoise bounding box) and Baltic Sea (blue bounding box).

III.1 Validation process when comparing to drifters and moored buoys

The individual in-situ observations have been matched to the closest satellite observations in space and time and all differences are computed as satellite – in-situ. Furthermore, to separate between the basins, match-ups are separated into the entire North Sea/Baltic Sea; with the Baltic Sea coverage only, defined as east of 9°E and, the North Sea coverage only, defined as west of 9°E, see Figure 2. The hourly nature of the new product allows for the separate validation of the day-time (06-19h) and night-time (19-06h) observations, performed for the entire domain.

In case no in-situ and/or satellite data is available, no validation takes place.

IV VALIDATION RESULTS

The following two sections present the results of the comparison of the L4 surface temperature product against drifters and moored buoys, respectively, for period May 1st 2021 to August 31st 2021. Conclusions on the results are presented in section IV.3, followed by a description of the impact of including NOAA21 data in the product.

IV.1 Validation when comparing to drifters

The results of the validation against drifters are summarised in Table 2 for the entire North Sea/Baltic Sea basin (row 1) and validation for the entire domain separated into day and night (row 2 and 3), the Baltic Sea (row 4) and the North Sea only (row 5). The regions are shown on Figure 2.

Validation metrics against drifters cannot be considered representative to the East of 9°E due to the very small amount (31 in total) of drifting buoys (see Table 2 below).

A cold bias for the L4 Diurnal product when compared against the drifting buoys is identified in all cases, manifested as negative mean bias values, persisting even when the statistics are calculated from separated data sets for day and night. This is consistent with the cold bias found against moored buoys for the corresponding foundation SST product SST_BAL_SST_L4_NRT_OBSERVATIONS_010_007_b; although slightly larger (-0.17 here vs -0.02) than the bias of the SST_BAL_SST_L4_NRT_OBSERVATIONS_010_007_b. The standard deviation and root mean square deviation (RMSD) values are similar to the numbers for the foundation temperature used to compute the hourly SST sub-skin fields. Such results indicate that the new product shows the stable performance, despite the short period used for the validation.

A small increase is seen in for all the metrics (bias, stddev and RMSD) on the 2nd decimal for the day-time period compared to night-time; although the number of observations for each period is balanced between both data samples (Table 2).

Table 2. Overall validation metrics for the SST_BAL_PHY_SUBSKIN_L4_NRT_010_034 against drifting buoys for the period May-August, 2021. The last two rows of the table include the validation statistics for the entire domain (North Sea/Baltic Sea) separated into day and night. The regions are shown in Figure 2.

Variable	Unit	Basin/Daily period	Period	Bias	Standard Deviation	RMSD	Number of match-ups
Temperature	°C	North Sea / Baltic Sea	All	-0.17	0.55	0.58	34690
			Day	-0.20	0.55	0.59	19869
			Night	-0.14	0.54	0.56	14821
		Baltic Sea	All	-0.52	0.69	0.86	31
		North Sea	All	-0.17	0.55	0.58	34659

Figure 3 shows the daily mean bias and daily standard deviation during the period May-August 2021, estimated between the hourly L4 analyses and drifting buoys. Both the mean bias and standard deviation are variable for the period of investigation although the mean bias becomes largely negative

and the standard deviation increases for the period end of June to July 10th and again around end of July. The validation results from the summer period is likely to be a high estimate for the full year statistics, as summer typically is the season with the highest uncertainty on the satellite surface temperature product.

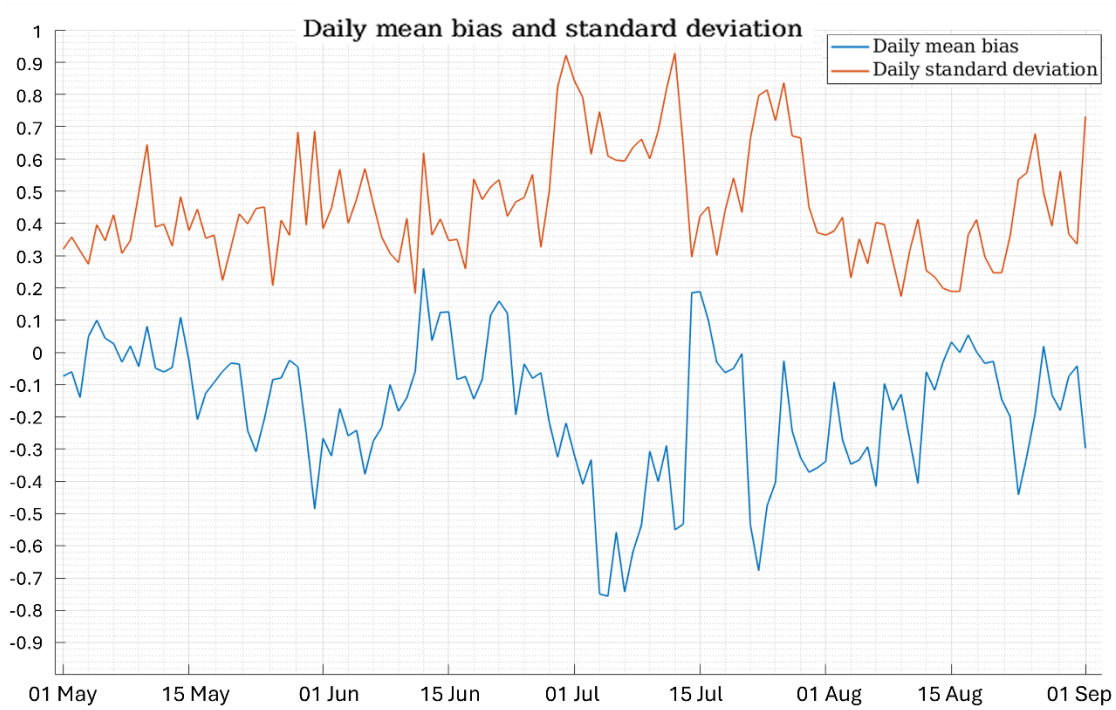

Figure 3. Daily validation statistics of the level 4 hourly analysis (May 1st 2021 to August 31st 2021) against drifting buoys: mean bias (blue), standard deviation (red).

IV.2 Validation results applied to moored buoys

The results of the validation against moored are summarised in Table 3 for the entire North Sea/Baltic Sea basin and the validation for the entire domain separated into day and night, the Baltic Sea and the North Sea only. Only one moored buoy was available in the North Sea for the period of investigation; hence no meaningful statistics could be derived.

Validation against the moored buoys indicates a slight warm bias for the L4 product, although negligible (0.05°), while the standard deviation and root mean square deviation (RMSD) are higher than what is reported against drifting buoys by approximately 0.2° to 0.3°C. This larger standard deviation is assumed to be due to the larger differences between SST sub-skin and the temperature at the depth of the sensor, which can be several meters on the moored buoys. The temperature sensor for the moored buoys can be at 0,5, 1m or 2m depths. Therefore, the validation results against moored buoys should be cautiously considered due to the large potential differences between the sub-skin SST and the temperature at 0,5-2 m depth. This difference is pronounced during the day and in spring and summer due to a larger thermocline in the upper part of ocean surface layer and can be seen in the results shown in Table 3, where the L4 product appears slightly warmer than the moored buoys.

While the bias is negligibly negative (-0.06°C) during day-time, a more pronounced positive bias (0.18°C) is found during night-time; indicating that the L4 product is warmer at night. Differences in the standard deviation and RMSD are about 0.1°C between day and night. Standard deviation and RMSD are larger

for day-time compared to nighttime, which likely reflects the fact that larger subskin vs. depth discrepancies are typically present during day than night.

Table 3. Overall validation metrics for the SST_BAL_PHY_SUBSKIN_L4_NRT_010_034 against moored buoys for the period May-August, 2021. First three rows include the validation statistics for the entire domain (North Sea/Baltic Sea) for all day, and separated into day and night, whereas the last two rows are for the separate two subregions, Baltic Sea and North Sea.

Variable	Unit	Basin/Daily period	Period	Bias	Standard Deviation	RMSD	Number of match-ups
Temperature	°C	North Sea/Baltic Sea	All	0.05	0.80	0.81	86317
			Day	-0.06	0.84	0.85	47289
			Night	0.18	0.73	0.75	39028
		Baltic Sea	All	0.05	0.80	0.81	86317
		North Sea	All	N/A	N/A	N/A	1

Figure 4 shows the daily mean bias and daily standard deviation during the period May-August 2021, estimated between the hourly L4 analyses and moored buoys. Both the mean bias and standard deviation are variable for the period of investigation with the mean bias being mostly positive, compared to what was found for the drifting buoys in Figure 3. The validation results from the summer period is likely to be a high estimate for the full year statistics, as summer typically is the season with the highest uncertainty on the satellite surface temperature product.

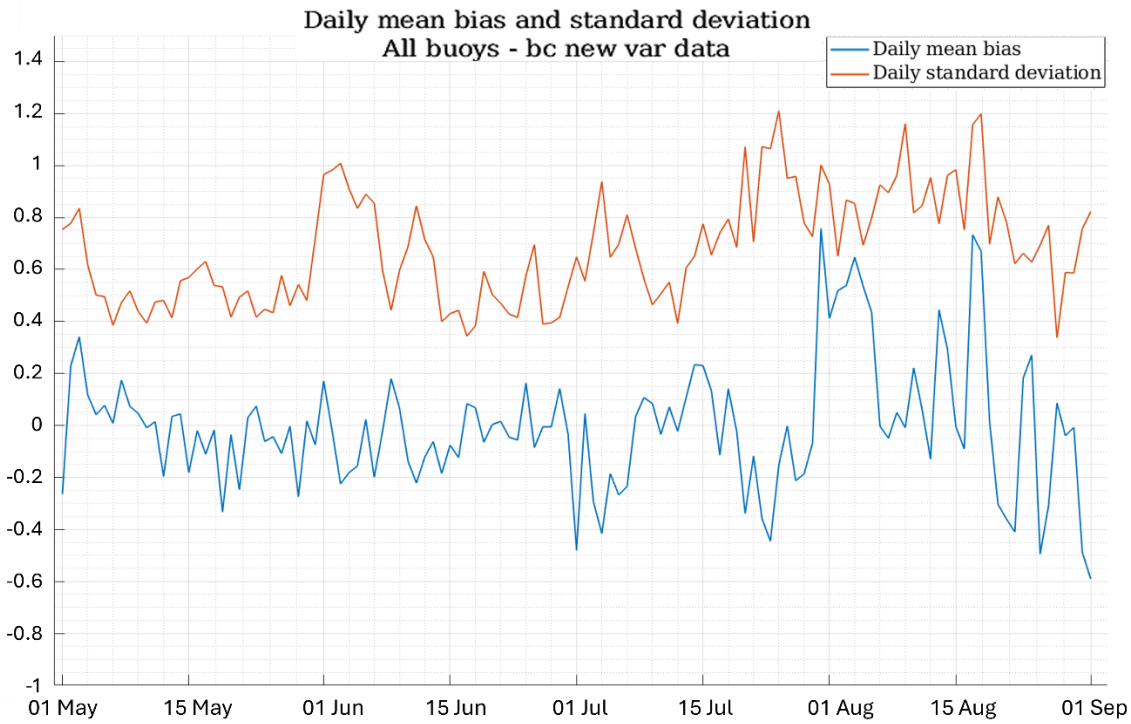

Figure 4. Daily validation statistics of the level 4 hourly analysis (May 1st, 2021 - August 31st, 2021) against moored buoys: mean bias (blue), standard deviation (red).

IV.3 Conclusion of the validation

Overall, validation statistics from all buoys shows good performance of the hourly analyses although the period of validation is limited. Figure 5 shows the daily mean bias and daily standard deviation during the period May-August 2021, estimated between the hourly L4 analyses and all buoys combined. Note both moored and drifting buoys have been combined to increase the daily number of validation points. With more available observations and a longer period of available hourly L4 analyses, more robust validation statistics will be deducted.

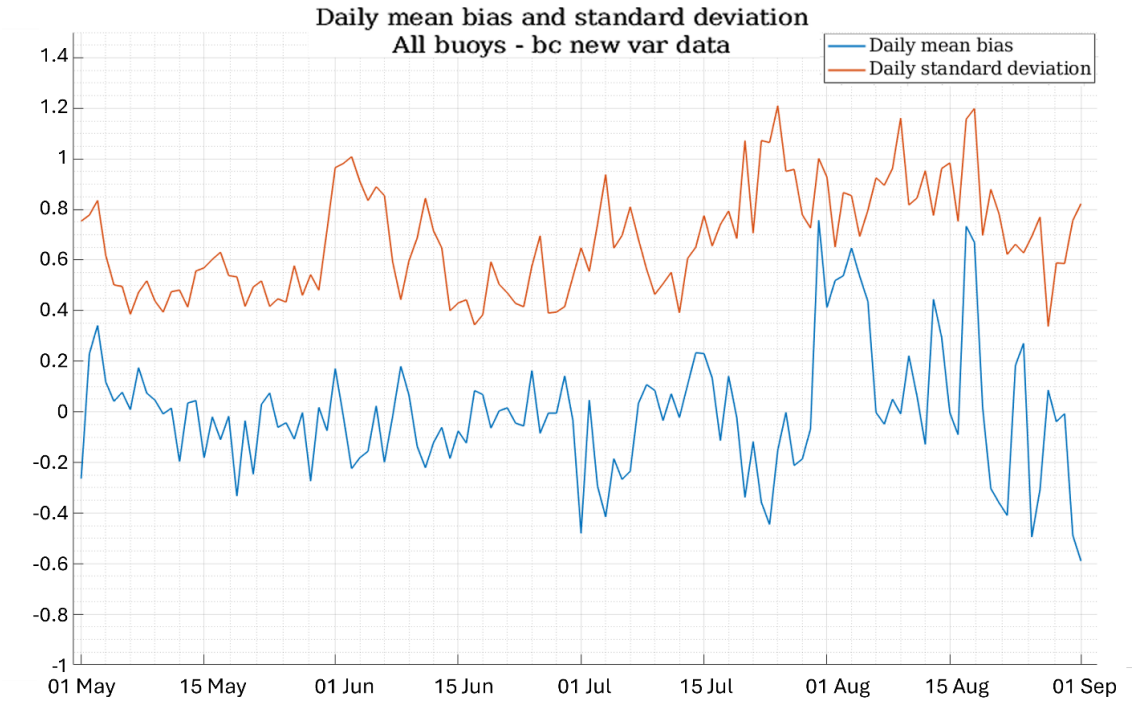

Figure 5. Daily validation statistics of the level 4 hourly analysis (May 1st, 2021 - August 31st, 2021) from the combination of drifting and moored buoys: mean bias (blue), standard deviation (red).

IV.4 Impact of adding VIIRS/NOAA21

In advance of the November 2025 Entry into the Service (EiS), SST observations from the VIIRS instrument on NOAA21 have been added to all NRT BAL products, including L3S, L4 and hourly BAL L4 SUBSKIN NRT (010-034). To assess the impact on the accuracy of the product of adding this new sensor, the period March-December 2023 was used as baseline and the BAL L3S and BAL L4 NRT products were re-processed to also include VIIRS on NOAA21. The hourly BAL L4 SUBSKIN NRT (010-034) product was not specifically re-processed as any impact on the L3S (010-032) and L4 NRT (010-007b) is assumed to be the same for the hourly BAL L4 SUBSKIN NRT (010-034). The results of the assessment are detailed in the Quality Information Document[12] CMEMS-SST-QUID-010-007-032. In summary no difference in the mean bias between the L4 NRT product 010-007b and in-situ observations from Argo floats, drifting and moored buoys is observed; while there is a slight reduction in the standard deviation when VIIRS on NOAA21 is included. The added benefit of including VIIRS/NOAA21 observations is mostly related to the fact that more observations are available that result in less gaps to be filled by the OI algorithm for the generation of the L4 product.

[12] See https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-007-032.pdf

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Date	Change/Event description	System version	Other
26/11/2025	Include VIIRS sensor on NOAA21.	1.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version. No major quality changes are recorded besides the inclusion of the VIIRS sensor on NOAA21 platform. From the analysis performed using the L3S and L4 NRT products (010-032 and 010-007b), it is demonstrated that although no major improvement in the validation with in-situ observations is achieved when including VIIRS on NOAA21, there is a slight reduction in the standard deviation and the added benefit of more observations available resulting in less gaps to be filled by the OI algorithm for the generation of the L4 product.

The results from the analysis above have been included in the QUID document[13], and are assumed to be valid for the hourly BAL L4 SUBSKIN NRT (010-034) product, due to the same nature of the processing system and the fact that the L4 NRT foundation product (010-007b) is used in the generation of the hourly BAL L4 SUBSKIN NRT product (010-034).

[13] See https://documentation.marine.copernicus.eu/QUID/CMEMS-SST-QUID-010-007-032.pdf

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